

UPP Mathematics Mock Test 1

Solution

Question 1. Evaluate the following improper integral:

$$\int_0^1 \frac{\ln(x)}{\sqrt{x}} dx.$$

Solution: Evaluate the improper integral:

$$\int_0^1 \frac{\ln(x)}{\sqrt{x}} dx = \lim_{a \rightarrow 0^+} \int_a^1 \frac{\ln x}{\sqrt{x}} dx.$$

Solution using integration by parts:

Let $u = \ln x$, hence $du = \frac{1}{x} dx$. Let $dv = \frac{1}{\sqrt{x}} dx$, then $v = 2\sqrt{x}$. Integration by parts formula is $\int u dv = uv - \int v du$.

Applying integration by parts:

$$\int \frac{\ln x}{\sqrt{x}} dx = 2\sqrt{x} \ln x - \int 2\sqrt{x} \cdot \frac{1}{x} dx = 2\sqrt{x} \ln x - 4\sqrt{x}$$

Thus, we have:

$$\lim_{a \rightarrow 0^+} (2\sqrt{x} \ln x - 4\sqrt{x}) \Big|_a^1 = \lim_{a \rightarrow 0^+} (0 - 4(1)) - (2\sqrt{a} \ln a - 4\sqrt{a})$$

As $a \rightarrow 0^+$, $4\sqrt{a} \rightarrow 0$ and

$$\lim_{a \rightarrow 0^+} (2\sqrt{a} \ln a) = \lim_{a \rightarrow 0^+} \frac{2 \ln(a)}{\frac{1}{\sqrt{a}}} = \lim_{a \rightarrow 0^+} \frac{2 \frac{1}{a}}{\frac{-1}{2} a^{-\frac{3}{2}}} = \lim_{a \rightarrow 0^+} (-4a^{\frac{1}{2}}) = 0$$

Hence, the final answer is -4 .

Question 2. Use Ratio Test to determine the convergence of the series

$$\sum_{k=1}^{\infty} \frac{k^4 \cdot 4^k}{(2k)!}$$

Solution: We apply the Ratio Test which examines the limit of the ratio of consecutive terms:

$$L = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}}{a_k} \right| = \lim_{k \rightarrow \infty} \left| \frac{(k+1)^4 \cdot 4^{k+1}}{(2(k+1))!} \cdot \frac{(2k)!}{k^4 \cdot 4^k} \right|$$

This simplifies to:

$$L = \lim_{k \rightarrow \infty} \frac{(k+1)^4}{k^4} \cdot \frac{4}{(2k+1)(2k+2)}$$

As $k \rightarrow \infty$,

$$\frac{(k+1)^4}{k^4} \rightarrow 1$$

and

$$\frac{4}{(2k+1)(2k+2)} \rightarrow 0$$

Thus L is 0 and by the Ratio Test, the series converges.

Question 3. Determine the interval of convergence of the power series:

$$\sum_{k=2}^{\infty} \frac{3^k}{k^2} (x-1)^k$$

Solution: We will first determine the radius of convergence using the Ratio Test. For the power series $a_k(x-1)^k$, the ratio of successive terms is:

$$L = \lim_{k \rightarrow \infty} \left| \frac{a_{k+1}(x-1)^{k+1}}{a_k(x-1)^k} \right| = \lim_{k \rightarrow \infty} \left| \frac{3^{k+1}}{(k+1)^2} \cdot \frac{k^2}{3^k} \right| |x-1|$$

Simplifying this expression, we get:

$$L = \lim_{k \rightarrow \infty} \left| \frac{3k^2}{(k+1)^2} \right| |x-1| = 3|x-1|$$

For the series to converge, we need $L < 1$, which gives us:

$$|x-1| < \frac{1}{3}$$

This implies the radius of convergence is $R = \frac{1}{3}$, and so the interval of convergence is $x \in (1 - \frac{1}{3}, 1 + \frac{1}{3})$.

To check the endpoints, we set $x = 1 - \frac{1}{3}$ and $x = 1 + \frac{1}{3}$ and compare to a p -series:

At $x = 1 - \frac{1}{3}$, the series becomes $\sum_{k=2}^{\infty} \frac{3^k}{k^2} \left(-\frac{1}{3}\right)^k$, which is a p -series with $p = 2$ that converges. At $x = 1 + \frac{1}{3}$, the series becomes $\sum_{k=2}^{\infty} \frac{3^k}{k^2} \left(\frac{1}{3}\right)^k$, which is a p -series with $p = 2$ that also converges.

Therefore, the interval of convergence is $[\frac{2}{3}, \frac{4}{3}]$.

Question 4.

The line ℓ_1 passes through point A with coordinates $(3, -1, 4)$ and is parallel to the vector $\begin{bmatrix} 4 \\ -2 \\ 5 \end{bmatrix}$.

The plane H contains the point C with coordinates $(1, -3, 2)$ and the line ℓ_1 . Determine:

- The standard equation of the plane H .
- Symmetric equation of the line ℓ_2 containing the point C and is perpendicular to the plane H .

Solution:

(a) Since the plane H contains the line ℓ_1 and the point C , the normal vector to the plane can be found using the direction vector of ℓ_1 and the vector from point A to point C , $\vec{AC}$. The vector $\vec{AC}$ is given by:

$$\vec{AC} = \begin{bmatrix} 1 - 3 \\ -3 - (-1) \\ 2 - 4 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix}$$

The normal vector $\vec{n}$ of the plane is then the cross product of $\vec{AC}$ and the direction vector of ℓ_1 , $\vec{d} = \begin{bmatrix} 4 \\ -2 \\ 5 \end{bmatrix}$:

$$\vec{n} = \vec{AC} \times \vec{d} = \begin{bmatrix} -2 \\ -2 \\ -2 \end{bmatrix} \times \begin{bmatrix} 4 \\ -2 \\ 5 \end{bmatrix} = \begin{bmatrix} -10 - 4 \\ -8 + 10 \\ 4 - (-8) \end{bmatrix} = \begin{bmatrix} -14 \\ 2 \\ 12 \end{bmatrix}$$

The standard equation of the plane H is given by $\vec{n} \cdot \begin{bmatrix} x - x_0 \\ y - y_0 \\ z - z_0 \end{bmatrix} = 0$:

$$-14(x - 1) + 2(y + 3) + 12(z - 2) = 0$$

Simplifying, we have:

$$-14x + 14 + 2y + 6 + 12z - 24 = 0$$

$$-14x + 2y + 12z - 4 = 0$$

The standard equation of the plane H is:

$$7x - y - 6z + 2 = 0$$

(b) To find the equation of the line ℓ_2 that is perpendicular to the plane H , we take the normal vector $\vec{n}$ as the direction vector.

$$\vec{n} = \begin{bmatrix} -14 \\ 2 \\ 12 \end{bmatrix}$$

The symmetric equations for the line ℓ_2 passing through point $C(1, -3, 2)$ are:

$$\frac{x-1}{-14} = \frac{y+3}{2} = \frac{z-2}{12}$$

Question 5.

Construct the standard equation of the tangent plane and parametric equations for the normal line to the surface given by $e^x \cos(y) + e^y \sin(z) - \ln(y+1) - e = 0$ at the point $(1, 0, \pi)$

Solution: Given the function:

$$f(x, y, z) = e^x \cos(y) + e^y \sin(z) - \ln(y+1) - e,$$

we find the gradient of f by computing its partial derivatives:

$$\frac{\partial f}{\partial x} = e^x \cos(y),$$

$$\frac{\partial f}{\partial y} = -e^x \sin(y) + e^y \sin(z) - \frac{1}{y+1},$$

$$\frac{\partial f}{\partial z} = e^y \cos(z).$$

we calculate the gradient at the point $(1, 0, \pi)$ to be:

$$\nabla f|_{(1,0,\pi)} = (e, -1, -1)$$

The equation for the tangent plane at this point can be written as:

$$e(x - 1) - (y - 0) - (z - \pi) = 0,$$

which simplifies to:

$$ex - y - z = e - \pi.$$

For the parametric equations of the normal line, we use the components of the gradient vector as direction ratios:

$$\begin{cases} x = 1 + et, \\ y = -t, \\ z = \pi - t. \end{cases}$$

Question 6.

Find and classify the critical points of the function $f(x, y) = x^2y + y^3 - 3x^2 - 3y^2 + 6$.

Solution: The function given is

$$f(x, y) = x^2y + y^3 - 3x^2 - 3y^2 + 6.$$

First, we find the first-order partial derivatives to obtain the gradient of f :

$$\frac{\partial f}{\partial x} = 2xy - 6x,$$

$$\frac{\partial f}{\partial y} = x^2 + 3y^2 - 6y.$$

Setting these partial derivatives to zero gives us the system of equations:

$$2xy - 6x = 0,$$

$$x^2 + 3y^2 - 6y = 0.$$

Solving this system yields the critical points $(0, 0)$, and $(0, 2)$.

Next, we use Second Partial Test by finding the discriminant of $f(x, y)$:

$$D_f = \begin{vmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{vmatrix} = \begin{vmatrix} 2y - 6 & 2x \\ 2x & 6y - 6 \end{vmatrix}$$

$$= (2y - 6)(6y - 6) - (2x)^2.$$

Evaluating D_f at each critical point:

For $(0, 0)$:

$$D_f|_{(0,0)} = \begin{vmatrix} -6 & 0 \\ 0 & -6 \end{vmatrix} = 36 > 0,$$

and since $\frac{\partial^2 f}{\partial x^2}|_{(0,0)} < 0$, point $(0, 0, 6)$ is a relative maximum point.

For $(0, 2)$:

$$D_f|_{(0,2)} = \begin{vmatrix} -2 & 0 \\ 0 & 6 \end{vmatrix} = -12 < 0,$$

which indicates that point $(0, 2, 2)$ is a saddle point.

Thus, we have one local maximum at $(0, 0, 6)$, and one saddle point at $(0, 2, 2)$.